IN21-S3-PL2301
Planning Theory

Road network Connectivity theory

Jaffna

Assignment 03

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Connectivity theory is a mathematical framework that studies the properties and relationships of networks composed of nodes and edges. It provides way to analysis complex systems and understand their structure, behaviour and dynamics. For this connectivity analysis I choose Jaffna town road network within a 25 km radius of the center of the Jaffna town area. For that I selected A class and prominently used AB class roads and their intersecting nodes 14 nodes within the Jaffna region.

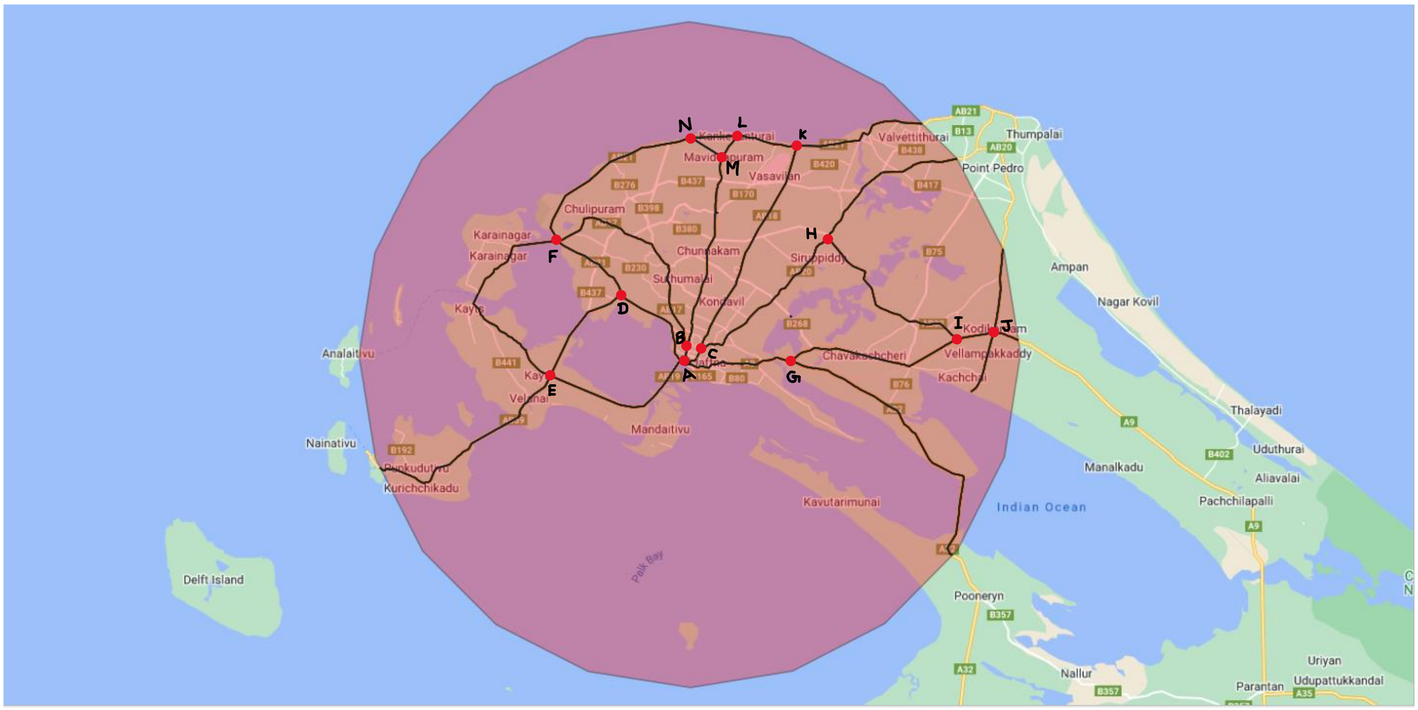

Figure 1 Node-Axis diagram of Jaffna region road network

After preparing this Node-Axis diagram, I prepared an interactive matrix with 14 points to calculate the relative connectivity.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
A	0	1	1	1	1	2	1	2	2	3	2	3	2	3
B	1	0	2	2	2	1	2	3	3	4	3	2	1	2
C	1	2	0	2	2	3	2	1	2	3	1	2	3	3
D	1	2	2	0	1	1	2	3	3	4	3	3	3	2
E	1	2	2	1	0	1	2	3	3	4	3	3	3	2
F	2	1	3	1	1	0	3	4	4	5	3	2	2	1
G	1	2	2	2	2	3	0	2	1	2	3	4	3	4
H	2	3	1	3	3	4	2	0	1	2	2	3	4	4
I	2	3	2	3	3	4	1	1	0	1	3	4	4	5
J	3	4	3	4	4	5	2	2	1	0	4	5	5	6
K	2	3	1	3	3	3	3	2	2	4	0	1	2	2
L	3	2	2	3	3	2	4	3	4	5	1	0	1	1
M	2	1	3	3	3	2	3	4	4	5	2	1	0	1
N	3	2	3	2	2	1	4	4	5	6	2	1	1	0

Figure 2 Interactive matrix table

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
A	0	1	1	1	1	2	1	2	2	3	2	3	2	3
B	1	0	2	2	2	1	2	3	3	4	3	2	1	2
C	1	2	0	2	2	3	2	1	2	3	1	2	3	3
D	1	2	2	0	1	1	2	3	3	4	3	3	3	2
E	1	2	2	1	0	1	2	3	3	4	3	3	3	2
F	2	1	3	1	1	0	3	4	4	5	3	2	2	1
G	1	2	2	2	2	3	0	2	1	2	3	4	3	4
H	2	3	1	3	3	4	2	0	1	2	2	3	4	4
I	2	3	2	3	3	4	1	1	0	1	3	4	4	5
J	3	4	3	4	4	5	2	2	1	0	4	5	5	6
K	2	3	1	3	3	3	3	2	2	4	0	1	2	2
L	3	2	2	3	3	2	4	3	4	5	1	0	1	1
M	2	1	3	3	3	2	3	4	4	5	2	1	0	1
N	3	2	3	2	2	1	4	4	5	6	2	1	1	0
	24	28	27	30	30	32	31	34	35	48	32	34	34	36

Figure 3 Shows relative connectivity values of each nodes.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	
A	0.00	0.04	0.04	0.03	0.03	0.06	0.03	0.06	0.06	0.06	0.06	0.09	0.06	0.08	0.71
B	0.04	0.00	0.07	0.07	0.07	0.03	0.06	0.09	0.09	0.08	0.09	0.06	0.03	0.06	0.84
C	0.04	0.07	0.00	0.07	0.07	0.09	0.06	0.03	0.06	0.06	0.03	0.06	0.09	0.08	0.82
D	0.04	0.07	0.07	0.00	0.03	0.03	0.06	0.09	0.09	0.08	0.09	0.09	0.09	0.06	0.90
E	0.04	0.07	0.07	0.03	0.00	0.03	0.06	0.09	0.09	0.08	0.09	0.09	0.09	0.06	0.90
F	0.08	0.04	0.11	0.03	0.03	0.00	0.10	0.12	0.11	0.10	0.09	0.06	0.06	0.03	0.97
G	0.04	0.07	0.07	0.07	0.07	0.09	0.00	0.06	0.03	0.04	0.09	0.12	0.09	0.11	0.95
H	0.08	0.11	0.04	0.10	0.10	0.13	0.06	0.00	0.03	0.04	0.06	0.09	0.12	0.11	1.07
I	0.08	0.11	0.07	0.10	0.10	0.13	0.03	0.03	0.00	0.02	0.09	0.12	0.12	0.14	1.14
J	0.13	0.14	0.11	0.13	0.13	0.16	0.06	0.06	0.03	0.00	0.13	0.15	0.15	0.17	1.54
K	0.08	0.11	0.04	0.10	0.10	0.09	0.10	0.06	0.06	0.08	0.00	0.03	0.06	0.06	0.96
L	0.13	0.07	0.07	0.10	0.10	0.06	0.13	0.09	0.11	0.10	0.03	0.00	0.03	0.03	1.06
M	0.08	0.04	0.11	0.10	0.10	0.06	0.10	0.12	0.11	0.10	0.06	0.03	0.00	0.03	1.05
N	0.13	0.07	0.11	0.07	0.07	0.03	0.13	0.12	0.14	0.13	0.06	0.03	0.03	0.00	1.11
	24	28	27	30	30	32	31	34	35	48	32	34	34	36	

Figure 4 Shows the normalized connectivity of every node.

Nodes	Row sum values	Connectivity values
A	0.71	1.42
B	0.84	1.19
C	0.82	1.23
D	0.90	1.11
E	0.90	1.11
F	0.97	1.03
G	0.95	1.05
H	1.07	0.94
I	1.14	0.88
J	1.54	0.65
K	0.96	1.04
L	1.06	0.95
M	1.05	0.96
N	1.11	0.90

Figure 5 Shows the connectivity level of each Node.

Nodes	Buildings within 100m radius
A	67
B	53
C	59
D	48
E	31
F	42
G	41
H	39
I	37
J	21
K	46
L	36
M	39
N	34

Figure 6 Shows the count of buildings in each node.

Figure 6 shows the number of buildings within 100m radius of every node. The accuracy of buildings density is a concern because of few limiting factors.

According to the relative connectivity values obtained, node A (1.42) recorded highest value in the Jaffna region. node C obtained 1.23 connectivity value next to node A. when comparing building density with connectivity values (figure 7), easily identified high connectivity value point has large number of buildings around it. According to this connectivity analysis, building density is comparatively high around the highest connectivity node like A.

Nodes	Connectivity values	Buildings within 100m radius
A	1.42	67
B	1.19	53
C	1.23	59
D	1.11	48
E	1.11	31
F	1.03	42
G	1.05	41
H	0.94	39
I	0.88	37
J	0.65	21
K	1.04	46
L	0.95	36
M	0.96	39
N	0.90	34

Figure 7 Comparison of connectivity values and buildings density.